I225E Statistical Signal Processing

3. Basics of Stochastic Process

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Stochastic processes

- Random walk
  (De Moivre-Laplace theorem, Stirling's formula)
- Wiener process
- Ergodicity

Random Walk

Random walk

[Problem]

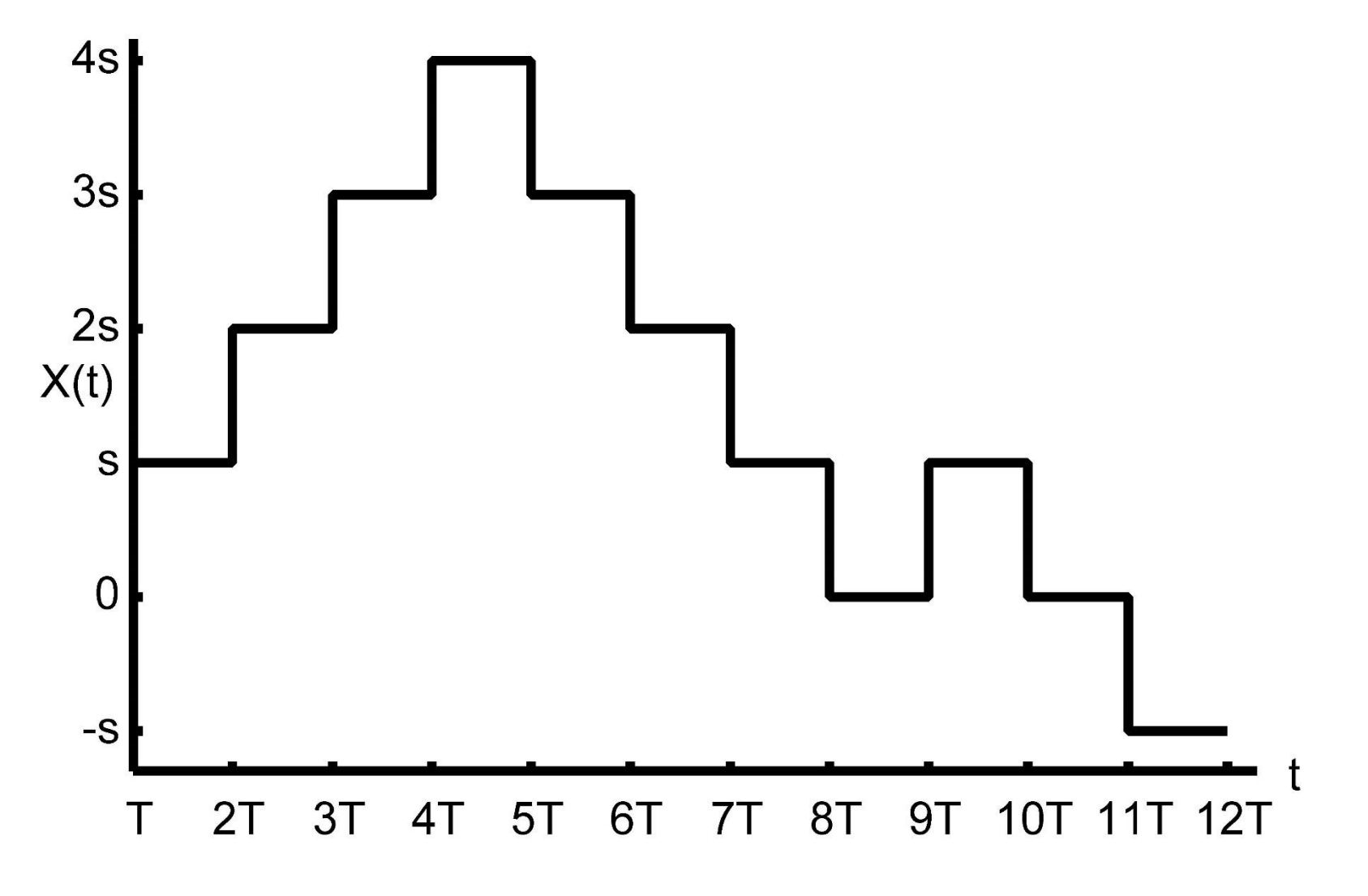
(a) Start at $t = 0$. Every time of T, throw a coin.

(b) If front face is up, proceed to right with s-step.

(c) If back face is up, proceed to left with s-step.

(d) Position at $t = nT$: $\boldsymbol{X}(\mathrm{t})$

Study the statistical quantities (mean, variance, and distribution function) of random variable $\boldsymbol{X}(t)$.


4s
3s
2s
X(t)
s
0
-s
t
T
2T
3T
4T
5T
6T
7T
8T
9T
10T
11T
12T

- If we suppose that, for the first n steps, front face was up for k times, and back face was up for $n-k$ times,

$$\boldsymbol{X}(nT) = ks - (n-k)s = ms$$

where $m = 2k - n, m = -n, -n+2, \cdots, n$

- Probability of obtaining front for k times among n trials is

$$P\{\boldsymbol{X}(nT) = ms\} = \binom{n}{k}\frac{1}{2^n} \quad \text{where } k = \frac{m+n}{2}$$

- Denoting the *i*th step by $\boldsymbol{X}(nT)$ can be described as $\boldsymbol{X}(nT) = \Delta\boldsymbol{X}_1 + \Delta\boldsymbol{X}_2 + \cdots + \Delta\boldsymbol{X}_n$. $\Delta\boldsymbol{X}_i$ $(= \pm s)$ is an independent random variable with $E\{\Delta\boldsymbol{X}_i\} = 0$ and $E\{\Delta\boldsymbol{X}_i^2\} = s^2$

$$E\{\boldsymbol{X}(nT)\} = nE\{\Delta\boldsymbol{X}_i\} = 0$$

$$E\{\boldsymbol{X}^2(nT)\} = nE\{\Delta\boldsymbol{X}_i^2\} = ns^2$$

- According to De Moivre-Laplace theorem,

  "If $npq \gg 1$, in $\sqrt{npq}$ neighborhood of $k = np$,

  $$\binom{n}{k} p^k q^{n-k} \cong \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(k-np)^2}{2npq}},$$

  Hence substituting $p = q = 0.5$, $m = 2k - n$,

  $$P\{\boldsymbol{X}(nT) = ms\} \cong \frac{1}{\sqrt{n\pi/2}} e^{-\frac{m^2}{2n}}$$ holds for $|m| \sim \sqrt{n}$.

  Therefore, $P\{\boldsymbol{X}(nT) \leq ms\} = \boldsymbol{\Phi}\left(\frac{m}{\sqrt{n}}\right)$ for $nT - T < t \leq T$

  where $\Phi(\cdot)$ represents distribution function of standard normal distribution $N(0,1)$. In addition, if $n_1 < n_2 \leq n_3 < n_4$, increments $\boldsymbol{X}(n_4T) - \boldsymbol{X}(n_3T)$ and $\boldsymbol{X}(n_2T) - \boldsymbol{X}(n_1T)$ are independent.

De Moivre-Laplace theorem: a derivation

$$\boxed{\binom{n}{k} p^k q^{n-k} \approx \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(k-np)^2}{2npq}}, \quad p+q=1}$$

[Outline of derivation] Using Stirling's formula for factorial, $n! \approx \sqrt{2\pi n}\left(n/e\right)^n$,

$$\binom{n}{k} p^k q^{n-k} \approx \sqrt{\frac{n}{2\pi k(n-k)}} \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k} .$$

Setting $k = npq + x\sqrt{npq}$ and expanding using a Taylor series $\ln(1+x) = x - x^2/2 + \cdots$,

$$\binom{n}{k} p^k q^{n-k} \approx \frac{1}{\sqrt{2\pi npq}} e^{-\frac{x^2}{2}} .$$

Exercise

Consider a simple symmetric discrete-time random walk $X[n]$ defined on an integer grid. The process starts at the origin, $X[0] = 0$. For each time step $n \geq 1$, the position is updated as:

$$X[n] = X[n-1] + W[n]$$

where $W[n]$ is an i.i.d. (independent and identically distributed) sequence of step variables with the following PMF:

$$P(W[n] = 1) = p$$
$$P(W[n] = -1) = 1 - p = q$$

(a) Derive the expression for the mean $E\big[X[n]\big]$ in terms of n, p, and q.

(b) Calculate the autocorrelation function $R_{XX}[n,m] = E\big[X[n]X[m]\big]$ for $n \geq m$.

(c) Determine if $X[n]$ is a Wide-Sense Stationary (WSS) process. Justify your answer.

Answer

(a) Mean $E\left[X[n]\right]$

The position $X[n]$ can be expressed as the sum of the individual steps:

$$X[n] = \sum_{i=1}^{n} W[i]$$

Using the linearity of expectation:

$$E\left[X[n]\right] = \sum_{i=1}^{n} E\left[W[i]\right]$$

Since $E\left[W[i]\right] = (1)(p) + (-1)(q) = p - q$:

$$E\left[X[n]\right] = n(p - q)$$

If the walk is symmetric ($p = 0.5$), the mean is 0.

(b) Autocorrelation $\boldsymbol{R_{XX}[n,m] = E\big[X[n]X[m]\big]}$ for $\boldsymbol{n \geq m}$.

We can decompose $X[n]$ into the sum of the walk up to time m and the steps taken after time m:

$$X[n] = X[m] + \sum_{i=m+1}^{n} W[i]$$

Substitute this into the autocorrelation formula:

$$E\big[X[n]X[m]\big] = E\left[\left(X[m] + \sum_{i=m+1}^{n} W[i]\right) X[m]\right]$$

$$E\big[X[n]X[m]\big] = E[X[m]^2] + E\left[X[m] \sum_{i=m+1}^{n} W[i]\right]$$

Because the steps $W[i]$ for $i > m$ are independent of $X[m]$, the second term becomes:

$$E\big[X[m]\big] \cdot E\left[\sum_{i=m+1}^{n} W[i]\right] = [m(p-q)] \cdot [(n-m)(p-q)]$$

(b) Autocorrelation $\boldsymbol{R_{XX}[n,m] = E\big[X[n]X[m]\big]}$ for $\boldsymbol{n \geq m}$. (Cont.)

For the first term, $E[X[m]^2]$ is the second moment.

Since $Var(X[m]) = E[X[m]^2] - \big(E\big[X[m]\big]\big)^2$,

and $Var(X[m]) = m \cdot Var(W) = m[1-(p-q)^2]$:

$$E[X[m]^2] = m(1-(p-q)^2) + m^2(p-q)^2$$

Combining and simplifying for the general case $n \geq m$:

$$R_{XX}[n,m] = m + (p-q)^2(nm - m)$$

(c) Justify if $X[n]$ is a Wide-Sense Stationary (WSS) process.

A process is WSS if:

- The mean $E\big[X[n]\big]$ is a constant (independent of n).

Mean: $E\big[X[n]\big] = n(p-q)$.

This depends on n unless $p = q = 0.5$.

Even if $p = q$, the mean is 0 (constant), but we must check the second condition.

- The autocorrelation $R_{XX}[n,m]$ depends only on the time lag $k = n - m$.

Variance/Autocorrelation: Even in the symmetric case, the variance $Var(X[n]) = R_{XX}[n,n] = n$. The variance grows linearly with time.

Conclusion: $X[n]$ is not WSS.

Wiener process

Consider a random walk in the limit of $n \to \infty$. We consider a limit $T \to 0$ under the condition of $s^2 = \alpha T$. Then $\boldsymbol{X}(t)$ becomes continuous-time continuous-state stochastic process

$$\boldsymbol{Y}(t) = \lim_{T \to 0} \boldsymbol{X}(t).$$

$\boldsymbol{Y}(t)$ is called ***Wiener process***.

Mean and Variance

According to the results of random walk,

$$E\{\boldsymbol{Y}(t)\} = 0$$

$$E\left\{\boldsymbol{Y}^{\boldsymbol{2}}(t)\right\} = ns^2 = \frac{ts^2}{T} = \alpha t.$$

- **Distribution function**: Substituting y= ms, $t = nT$ into distribution function of random walk,

$$P\{\boldsymbol{Y}(t) \leq y\} = \boldsymbol{\Phi}\left(\frac{m}{\sqrt{n}}\right) = \boldsymbol{\Phi}\left(\frac{y/s}{\sqrt{t/T}}\right) = \boldsymbol{\Phi}\left(\frac{y}{\sqrt{\alpha t}}\right)$$

Hence, probability density of $\boldsymbol{Y}(t)$ is distributed normally as $N(0, \alpha t)$.

$$f_{\boldsymbol{Y}}(y, t) = \frac{1}{\sqrt{2\pi\alpha t}} e^{-\frac{y^2}{2\alpha t}}$$

- **Autocorrelation**:

$$R_{\boldsymbol{YY}}(t_1, t_2) = \alpha \min(t_1, t_2)$$

- **Increment**: if $t_1 < t_2 \leq t_3 < t_4$, increments $\boldsymbol{Y}(t_4) - \boldsymbol{Y}(t_3)$ and $\boldsymbol{Y}(t_2) - \boldsymbol{Y}(t_1)$ are intendant.

Exercise

A Wiener process $W(t)$ is a continuous-time stochastic process that serves as the mathematical model for Brownian motion. It is defined by the following properties:

1. $W(0) = 0$.
2. $W(t)$ has independent increments.
3. For $t > s$, the increment $W(t) - W(s)$ is normally distributed with mean 0 and variance $t - s$.
4. $W(t)$ has continuous paths.

Questions:

- Find the probability that the process is greater than 2 at time $t = 4$. That is, calculate $P(W(4) > 2)$.
- Derive the autocovariance function $C_W(t, s) = E[W(t)W(s)]$ for any $t, s \geq 0$.

Answer

- **Find the probability that the process is greater than 2 at time $t = 4$. That is, calculate $P(W(4) > 2)$.**

From the definition, $W(4) - W(0) \sim \mathcal{N}(0,4-0)$.

Since $W(0) = 0$:

$$W(4) \sim \mathcal{N}(0,4)$$

The standard deviation is $\sigma = \sqrt{4} = 2$. We want to find $P(W(4) > 2)$. We convert this to a standard normal variable $Z \sim \mathcal{N}(0,1)$:

$$P(W(4) > 2) = P\left(\frac{W(4) - 0}{2} > \frac{2 - 0}{2}\right) = P(Z > 1)$$

Using the standard normal table or Q-function:

$$P(Z > 1) \approx 1 - 0.8413 = 0.1587$$

Answer

- **Derive the autocovariance function $C_W(t,s) = E[W(t)W(s)]$ for any $t, s \geq 0$.**

Assume without loss of generality that $s \leq t$. We can write $W(t)$ as:

$$W(t) = W(s) + \big(W(t) - W(s)\big)$$

The covariance is:

$$E[W(t)W(s)] = E\left[\Big(W(s) + \big(W(t) - W(s)\big)\Big)W(s)\right]$$

$$E[W(t)W(s)] = E[W(s)^2] + E\big[\big(W(t) - W(s)\big)W(s)\big]$$

Since $W(s) \sim \mathcal{N}(0, s)$, then

$$E[W(s)^2] = Var\big(W(s)\big) + (E[W(s)])^2 = s + 0 = s$$

Because the Wiener process has independent increments, the increment $\big(W(t) - W(s)\big)$ is independent of the value at $W(s)$. So, the second term is:

$$E[W(t) - W(s)] \cdot E[W(s)] = 0 \cdot 0 = 0$$

$$\therefore C_W(t,s) = \min(t,s)$$

Generalized random walk

- In random walk, we suppose that probability of obtaining front face is p, whereas probability of obtaining back is $q = 1 - p$. Then,

$$\boldsymbol{X}(t) = \sum_{k=1}^{n} c_k U(t - kT) \quad \text{for} \quad (n-1)T < t \leq T$$

where c_k is a random number, which takes value of s with probability p and takes a value of $-s$ with probability q and

$$U(t) = 0 \ (t < 0) \text{ and } U(t) = 1 \ (t \geq 0).$$

$\boldsymbol{X}(t)$ is called ***generalized random walk***.

Generalized random walk

Using the following properties of binominal distribution:

$$E\{c_k\} = (p-q)s$$

$$E\{c_k^2\} = s^2, \qquad Var(c_k^2) = 4pqs^2$$

- **Mean and Variance:**

$$E\{\boldsymbol{X}(t)\} = n(p-q)s$$

$$Var(\boldsymbol{X}(t)) = 4npqs^2$$

- **Distribution function:**

For large *n*, $X(t)$ is normally distributed with

$$E\{\boldsymbol{X}(t)\} \cong \frac{t}{T}(p-q)s$$

$$Var(\boldsymbol{X}(t)) \cong \frac{4t}{T}4pqs^2$$

Ergodic property

- **Problem:**

  Consider an estimation of statistical quantity of $\boldsymbol{X}(t)$ such as its mean.

  $\eta(t) = E\{\boldsymbol{X}(t)\}$, from real data.

- **Method:**

  Given n samples $\boldsymbol{X}(t, \omega_i)$ $(i = 1, 2, \cdots, n)$, average is obtained as follows.

$$\hat{\eta}(t) = \frac{1}{n}\sum_{i=1}^{n} X(t, \omega_i)$$

- **Practical Problem:**

  It is rare to have some many samples. In most cases, only a single time series $\boldsymbol{X}(t)$ is given.

- **Non-stationary:**

  If $\boldsymbol{X}(t)$ is non-stationary and mean $E\{\boldsymbol{X}(t)\}$ is a function of t, estimation is impossible.

  However, if $\boldsymbol{X}(t)$ is stationary, time-average, computed as

$$\eta_T = \frac{1}{2T}\int_{-T}^{T} \boldsymbol{X}(t)dt$$

  becomes

$$\eta_T \to E\{\boldsymbol{X}\} \text{ as } T \to \infty$$

- Ergodic property implies **time-average equals to ensemble average.**

Mean-ergodic process

Problem:

Given a stationary real process $\boldsymbol{X}(t)$, compute its average $\eta = E\{\boldsymbol{X}(t)\}$. Define a time average over a duration of $2T$

$$\eta_T = \frac{1}{2T}\int_{-T}^{T} \boldsymbol{X}(t)dt$$

as a new random variable, average of η_T is

$$E\{\eta_T\} = \frac{1}{2T}\int_{-T}^{T} E\{\boldsymbol{X}(t)\}dt = \eta$$

If the variance has a property of $\sigma_T^2 \to 0$ in the limit of $T \to \infty$ time average converges to the true average. Namely,

$$P(\eta_T = \eta) \to 1$$

$\boldsymbol{X}(t)$ is called ***Mean-ergodic process***.

Slutsky theorem:

- If $\frac{1}{T}\int_0^T C(\tau)d\tau \to 0$ as $T \to \infty$, $\boldsymbol{X}(t)$ is a mean-ergodic process.
- Sufficient condition (a): $\int_0^\infty C(\tau)d\tau < \infty$
- Sufficient condition (b): For t $\to \infty$, $C(\tau) \to 0$

$$\mathrm{E}\left[(\eta_T-\eta)^2\right] = \frac{1}{(2T)^2}\int_{-T}^{T} dt \int_{-T}^{T} dt' \mathrm{E}\left[\left(x(t)-\eta\right)\left(x(t')-\eta\right)\right]$$

$$= \frac{2}{(2T)^2}\int_{-2T}^{2T} du \int_{-2T}^{2T} d\tau C(\tau)$$

$$= \frac{2}{T}\int_{-2T}^{2T} d\tau C(\tau)$$